# Determination of the Neutrino Time of Flight from the NuMI Target to the NOvA Detectors: Coordinate Systems, Distances and Timing Offsets

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**Abstract**

This note derives, in full, the straight-line distances between the NuMI target, the NOvA Near Detector (ND), the NOvA Far Detector (FD) at Ash River and the MINOS Far Detector at Soudan, and the corresponding neutrino times of flight in SI units and in 64 MHz NOvA clock ticks. Five coordinate systems are used and written out explicitly: a spherical Earth, WGS84 geodetic coordinates and the Earth-centred Earth-fixed (ECEF) frame, the Fermilab Site Coordinate System (a double stereographic projection), the NuMI beamline system with origin MCZERO in which the Fermilab alignment group surveyed both NOvA upstream-face centres, and a beam-axis frame used for off-axis and opening angles. The target is taken from the 1998 NuMI Technical Design Report beam sheet; the detector endpoints are the surveyed face centres. Two further sites, points 100 ft below the surfaces of Lake Vermilion and of Lake Superior at Two Harbors, are carried through the same chain. Every correction between the reference model and the final result is tabulated, together with chord geometry, the Main Injector timing shift, the trigger window, relativistic and instrumental terms, sensitivities and an error budget. The headline result is a flight time from the ND face to the FD face of 2,700.250552 µs, i.e. 172,816.035 ticks, and a total offset including the 141-turn Main Injector shift of 272,736 ticks. The 2013 Mathematica notebook and the 2014 note [1] are reproduced to $10^{-9}$ and their superseded geometries are compared with the surveyed result in an appendix. All computations are performed by the `gps-tof` package [2]; the tables of this note are generated from its JSON report.

## Contents

**1 Introduction** **2**

**2 Reference frames and transformations** **3**

2.1 Angles . . . . . 3

2.2 Geodetic coordinates and the ECEF frame . . . . . 3

2.3 Spherical-Earth model . . . . . 4

2.4 Local frame and antenna-to-detector offsets . . . . . 4

2.5 Fermilab Site Coordinate System . . . . . 4
2.6 NuMI beamline system (origin MCZERO) and the beam-axis frame . . . . . 5

**3 Input positions** **6**
3.1 NuMI target, MCZERO and the design beam axis . . . . . 6
3.2 Surveyed upstream-face centres of the NOvA detectors . . . . . 6
3.3 GPS antenna fixes, MINOS far detector and other inputs . . . . . 6
3.4 Lake sites . . . . . 7
3.5 Converted positions . . . . . 8

**4 Distances** **8**
4.1 Chords in the spherical and ellipsoidal models . . . . . 8
4.2 Correction ledger . . . . . 9
4.3 Chord geometry . . . . . 9

**5 Time of flight and timing offsets** **11**
5.1 Flight time and clock ticks . . . . . 11
5.2 Main Injector timing shift . . . . . 12
5.3 Combined offset and trigger window . . . . . 12
5.4 Effects outside the geometric flight time . . . . . 12

**6 Angles relative to the beamline** **13**
6.1 Beam axis and off-axis positions . . . . . 13
6.2 Opening angles . . . . . 13
6.3 Near Detector face edges . . . . . 14

**7 Lake sites: Lake Vermilion and Lake Superior at Two Harbors** **14**

**8 Validation** **15**

**9 Sensitivities and error budget** **18**

**10 Summary** **18**

**A History and reproduction of the earlier calculations** **20**

**B Software and reproducibility** **21**

# 1 Introduction

The NOvA data acquisition issues a "beam" trigger from the NuMI spill time $t_0$ delayed by the neutrino flight time $t_{\mathrm{delay}}$ from the beam source to the far detector. The spill is nominally 10 µs long; the DAQ discretises data into 50 µs microslices and the event builder extracts whole microslices, so

the trigger window is the half-open interval [1]

$$\Delta T_{\mathrm{trig}} = \left[\, t_0 + t_{\mathrm{delay}} - 20\,\mu\mathrm{s},\; t_0 + t_{\mathrm{delay}} + 30\,\mu\mathrm{s}\,\right), \tag{1}$$

which always overlaps exactly two closed microslices. $t_{\mathrm{delay}}$ has to be set a priori from the geometry.

The 2013 Mathematica notebook [3] computed the chord between the MINOS surface-building GPS antenna, lowered by 300 m, and an Ash River antenna. The 2014 note NOvA-doc-11860 [1] documented the antenna fixes and specified antenna-to-detector offsets. Neither contained the NuMI target, and the horizontal antenna-to-detector vectors were left to be determined. This note (i) derives the target position from the NuMI Technical Design Report [4] through the Fermilab site projection [5], (ii) records the surveyed upstream-face centres of both NOvA detectors [6, 7] in the NuMI beamline system and converts them to geodetic and ECEF coordinates, (iii) writes out every coordinate transformation used, and (iv) tabulates the distances, flight times, angles and uncertainties. Figures 1–4 are schematic views of the geometry; Sec. 7 and Figs. 5–7 treat two additional lake sites.

# 2 Reference frames and transformations

## 2.1 Angles

Survey records give angles in sexagesimal form. Degrees, minutes and seconds are converted to radians by

$$\mathrm{LC}(d, m, s) = \frac{\pi}{180}\left(d + \frac{m + s/60}{60}\right), \tag{2}$$

with the hemisphere letter supplying the sign. Longitudes are east-positive throughout; the 2013 notebook entered west longitudes as positive numbers, which leaves distances unchanged (a reflection is an isometry) but flips the sign of rotation-dependent terms such as the Sagnac correction (Sec. 5.4).

## 2.2 Geodetic coordinates and the ECEF frame

The reference ellipsoid is WGS84 [8], with semi-major axis $a$, semi-minor axis $b = a(1 - f)$ and first eccentricity $e^2 = (a^2 - b^2)/a^2$:

$$\begin{aligned} a &= 6{,}378{,}137\,\mathrm{m}, & 1/f &= 298.257223563, \\ b &= 6{,}356{,}752.31424518\,\mathrm{m}, & e^2 &= 6.69437999 \times 10^{-3}. \end{aligned} \tag{3}$$

The Fermilab site survey is in NAD83 on the GRS80 ellipsoid ($1/f = 298.257222101$), which differs from WGS84 by 0.1 mm in $b$; the two are treated as coincident, as the Fermilab coordinate-system document itself does [5]. A point with geodetic latitude $\varphi$, longitude $\lambda$ and ellipsoidal height $h$ maps to Earth-centred Earth-fixed Cartesian coordinates by

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2\varphi}}, \qquad \begin{aligned} X &= (N + h)\cos\varphi\cos\lambda, \\ Y &= (N + h)\cos\varphi\sin\lambda, \\ Z &= \big(N(1 - e^2) + h\big)\sin\varphi. \end{aligned} \tag{4}$$

$N$ is the prime-vertical radius of curvature. The $X$ axis passes through the intersection of the equator and the prime meridian, $Z$ through the pole. The inverse transformation is evaluated iteratively

to $10^{-14}$ rad. The height $h$ in Eq. (4) is the height above the ellipsoid. A GPS receiver's `$GPGGA` altitude is orthometric (above mean sea level), $H$; the same sentence gives the geoid separation $N_g$ and $h = H + N_g$. Each input position in this note carries an explicit datum, and the conversion appears as a line in the correction ledger.

### 2.3 Spherical-Earth model

As a reference model, used only to show the size of the ellipsoidal correction, the Earth is taken as a sphere of radius $R = a$ and

$$\vec{r} = (R + h)\,(\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \tag{5}$$

A sphere of equatorial radius places both sites about 10 km too far from the Earth's centre, and since the chord scales with the radius the Fermilab–Ash River chord comes out 823 m (176 ticks) too long (Sec. 4.1).

### 2.4 Local frame and antenna-to-detector offsets

At a point $(\varphi, \lambda)$ the local East, North and Up unit vectors in ECEF are

$$\begin{aligned} \hat{E} &= (-\sin\lambda,\ \cos\lambda,\ 0), \\ \hat{N} &= (-\sin\varphi\cos\lambda,\ -\sin\varphi\sin\lambda,\ \cos\varphi), \\ \hat{U} &= (\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \end{aligned} \tag{6}$$

A displacement $(d_E, d_N, d_U)$ from a surveyed point (an antenna) to the physics endpoint (a detector face) is added as $\delta\vec{p} = d_E\hat{E} + d_N\hat{N} + d_U\hat{U}$. With the ellipsoid normal as $\hat{U}$ a purely vertical offset is identical to changing $h$ by $d_U$. The 2014 note describes the plane tangent to a *sphere*, $\hat{U} = \vec{p}/|\vec{p}|$, which differs from the geodetic normal by the geodetic–geocentric latitude difference (0.19° at these sites); both are implemented and the difference is reported as a sensitivity.

### 2.5 Fermilab Site Coordinate System

Beam elements in the NuMI Technical Design Report are given in the Fermilab site system, historically "DUSAF" coordinates: $x$ east, $y$ north, $z$ elevation, in US survey feet (1 ft = 1200/3937 m). Beams-doc-1148 [5] defines the modern realisation, the FSCS, as a double stereographic projection of NAD83/GRS80 geodetic coordinates: the ellipsoid is mapped conformally to a sphere and the sphere projected stereographically to the plane. This is the Oblique Stereographic projection, EPSG method 9809 [9]. The defining parameters are listed in Table 1.

With $e$ the first eccentricity and $\varphi_0, \lambda_0$ the origin, the projection constants are

$$\begin{aligned} &\rho_0 = \frac{a(1-e^2)}{(1-e^2\sin^2\varphi_0)^{3/2}}, \qquad \nu_0 = \frac{a}{\sqrt{1-e^2\sin^2\varphi_0}}, \qquad R = \sqrt{\rho_0\nu_0}, \\ &n = \sqrt{1 + \frac{e^2\cos^4\varphi_0}{1-e^2}}, \qquad w_1 = \left[\frac{1+\sin\varphi_0}{1-\sin\varphi_0}\left(\frac{1-e\sin\varphi_0}{1+e\sin\varphi_0}\right)^{e}\right]^{n}, \qquad \sin\chi_0' = \frac{w_1-1}{w_1+1}, \\ &c = \frac{(n+\sin\varphi_0)(1-\sin\chi_0')}{(n-\sin\varphi_0)(1+\sin\chi_0')}, \qquad w_2 = c\,w_1, \qquad \chi_0 = \arcsin\frac{w_2-1}{w_2+1}, \end{aligned} \tag{7}$$

Table 1: Parameters of the Fermilab Site Coordinate System [5].

| | |
|---|---|
| origin (monument A0) latitude $\varphi_0$ | N 41°50′14.312704″ |
| origin longitude $\lambda_0$ | W 88°15′41.143123″ |
| origin ellipsoidal height $h_0$ | 186.49880 m |
| geodetic azimuth of the $+Y$ axis at A0, $\alpha$ | 38°16′48.01429″ |
| false easting, false northing | 100,000 ft, 100,000 ft |
| scale factor at A0, 720 ft above the DUSAF datum, $F_0$ | 1.000029251309483 |
| DUSAF $\rightarrow$ NAVD88 height correction, $\delta H$ | $-0.17308$ m |
| geoid height at A0 (GEOID93), $N_0$ | $-32.78456$ m |
| ellipsoid | GRS80 ($a = 6{,}378{,}137$ m, $1/f = 298.257222101$) |

and a point $(\varphi, \lambda)$ projects to the conformal sphere and then to the plane by

$$\begin{aligned} &\Lambda = n(\lambda - \lambda_0) + \lambda_0, \qquad w = c\left[\frac{1+\sin\varphi}{1-\sin\varphi}\left(\frac{1-e\sin\varphi}{1+e\sin\varphi}\right)^{e}\right]^{n}, \qquad \chi = \arcsin\frac{w-1}{w+1}, \\ &B = 1 + \sin\chi\sin\chi_0 + \cos\chi\cos\chi_0\cos(\Lambda-\lambda_0), \\ &E = \frac{2R\cos\chi\sin(\Lambda-\lambda_0)}{B}, \qquad N = \frac{2R\left[\sin\chi\cos\chi_0 - \cos\chi\sin\chi_0\cos(\Lambda-\lambda_0)\right]}{B}. \end{aligned} \tag{8}$$

The FSCS axes are rotated so that $+Y$ points along the geodetic azimuth $\alpha$, rescaled to true scale at 720 ft above the DUSAF datum, and offset by the false coordinates:

$$X_{\mathrm{FSCS}} = \frac{F_0\,(E\cos\alpha - N\sin\alpha)}{1\,\mathrm{ft}} + 100{,}000\,\mathrm{ft}, \qquad Y_{\mathrm{FSCS}} = \frac{F_0\,(N\cos\alpha + E\sin\alpha)}{1\,\mathrm{ft}} + 100{,}000\,\mathrm{ft}. \tag{9}$$

Heights relate as $h = H_{\mathrm{DUSAF}} + \delta H + N_g$ with $N_g$ taken as the A0 value across the site (the GEOID93 slope between A0 and the Main Injector centroid is 1.3 cm). The inverse of Eq. (8) is the EPSG inverse (iteration on the isometric latitude); the implementation reproduces the EPSG worked example (Amersfoort) to 2 mm and the CFMI control point of Beams-doc-1148 to 0.1 mm horizontally. A beam-sheet bearing $\beta$ (degrees counter-clockwise from site East) corresponds to the geodetic azimuth $(90^\circ - \beta) + \alpha$ at A0; the NuMI final bearing 152.18332° gives 336.097°, against 336.098° from the 1999 Fermilab–Soudan geodetic survey [10, 11].

### 2.6 NuMI beamline system (origin MCZERO) and the beam-axis frame

The Fermilab alignment group reports NuMI positions in a Cartesian beamline system: origin MCZERO; $X$ positive to beam right (eastward) looking downstream; $Y$ the stationing, positive downstream; $Z$ positive upward and orthogonal to the beam [6, 7]. The same construction serves as the beam-axis frame for off-axis angles. Given an origin $O$ and a second point $T$ on the line, with ECEF positions $\vec{p}_O$, $\vec{p}_T$ and the ellipsoid normal $\hat{n}_O$ at the origin,

$$\hat{u} = \frac{\vec{p}_T - \vec{p}_O}{|\vec{p}_T - \vec{p}_O|}, \qquad \hat{h} = \frac{\hat{u}\times\hat{n}_O}{|\hat{u}\times\hat{n}_O|}, \qquad \hat{v} = \hat{h}\times\hat{u}. \tag{10}$$

$\hat{u}$ is along the line, $\hat{h}$ to the right looking downstream (horizontal at the origin), $\hat{v}$ up, orthogonal to the line and in the vertical plane through $\hat{n}_O$. A point $P$ with $\vec{r} = \vec{p}_P - \vec{p}_O$ has beamline coordinates and off-axis quantities

$$Y = \vec{r}\cdot\hat{u}, \qquad X = \vec{r}\cdot\hat{h}, \qquad Z = \vec{r}\cdot\hat{v}, \qquad \theta_{\mathrm{off}} = \arctan\frac{\sqrt{X^2+Z^2}}{Y}, \qquad \psi_{\mathrm{clock}} = \mathrm{atan2}(X, Z), \tag{11}$$

where $\psi_{\text{clock}}$ is 0° above the axis and 90° to the right; the inverse is $\vec{p}_P = \vec{p}_O + Y\hat{u} + X\hat{h} + Z\hat{v}$. The angle between a chord $\vec{d}$ and the axis is $\angle(\vec{d}, \hat{u})$ with components atan2($\vec{d}\cdot\hat{h}, \vec{d}\cdot\hat{u}$) (right) and atan2($\vec{d}\cdot\hat{v}, \vec{d}\cdot\hat{u}$) (up). The opening angle at a vertex $V$ between the chords to $A$ and $B$ is $\angle(\vec{p}_A - \vec{p}_V, \vec{p}_B - \vec{p}_V)$, decomposed into the azimuth and elevation differences of the two chords at $V$; the transverse distance of $A$ from the line $VB$ is $|\vec{p}_A - \vec{p}_V| \sin\angle$. Angles between vectors are evaluated as atan2($|\vec{a}\times\vec{b}|, \vec{a}\cdot\vec{b}$) for numerical stability at the microradian level.

**Realisation for NuMI.** Two elements of the beamline system are not stated in the survey notes and were established here. (1) MCZERO lies at the upstream end of horn 1, between the target exit and the horn; it is taken as the horn-1 upstream face of the 1998 TDR beam sheet (row 0089, 1.5 m downstream of the target entrance). The as-built point is not public, so the origin is good to about 1 m along the beam, which is 0.2 ticks on an 810 km chord and nothing transverse. (2) The direction of the $Y$ axis. Two candidate lines were tested: the 1998 design axis (target → decay-pipe end) and the as-built aiming line (MCZERO → MINOS FD midpoint). The FD survey note gives an independent NAVD88 elevation of the face centre, $H = 1{,}209.1058$ ft $= 368.535$ m, i.e. $h \simeq 336.07$ m with the Ash River geoid separation $N_g = -32.462$ m of the receiver. The aiming line reproduces it to 1.5 m (0.3 m with the GEOID18 separation of −31.221 m instead of the receiver's value); the design axis misses it by 17 m. The aiming line is therefore the surveyors' $Y$ axis and is used for both survey points (Sec. 8).

**Detector faces.** A face of half-width $w$ and half-height $w'$ centred at $(X, Y, Z)$ has edge points $(X \mp w, Y, Z)$ (left, right) and $(X, Y, Z \pm w')$ (top, bottom) in the same frame; the face is taken perpendicular to the beamline rather than to the detector's own axis, a 3 cm difference for the ND. The angle the face subtends at the axis origin is $\angle(\vec{r}_{\text{left}}, \vec{r}_{\text{right}})$.

# 3 Input positions

## 3.1 NuMI target, MCZERO and the design beam axis

The NuMI Technical Design Report [4], Appendix A beam sheet "horn2 ces", lists every beamline element in DUSAF coordinates (entrance of the device). The rows used are the target entrance (0087, `HALTA TAR-2`), the horn-1 entrance (0089, `IHHRN1`, the MCZERO proxy) and the decay-pipe end (0092, `DENCU`); the sheet's final bearing is 152.18332° (ccw from site East) and pitch −3.336°. These are design positions; as-built target positions in the LE and ME configurations differ at the metre level.

## 3.2 Surveyed upstream-face centres of the NOvA detectors

The definitive endpoints are the upstream-face centres surveyed by the Fermilab Survey, Alignment and Geodesy group [6, 7], Table 2. The FD point is 7.5 in from the intersection of the rail midline with the wall (the bookend) and vertically at the measured highest floor point (1,181.8175 ft NAVD88) plus the 23.25 in pallet and the 304.21 in half-plane, $H = 1{,}209.1058$ ft. The ND point uses the $3 \times 3$-module width and height of 158.125 in and the tunnel cross-section drawing 489389.

## 3.3 GPS antenna fixes, MINOS far detector and other inputs

Table 3 lists the remaining inputs: the MINOS surface-building mast antenna and the three Ash River antennas from the 2014 note [1] (decoded from the TDU receivers' `$GPGGA` sentences; the Table 2

Table 2: Surveyed upstream-face centres in the NuMI beamline system (origin MCZERO; $X$ right, $Y$ stationing, $Z$ up) [6, 7]. Frame as realised in this note.

| point | $X$ (m) | $Y$ (m) | $Z$ (m) | frame |
|---|---|---|---|---|
| NOvA ND upstream face centre (Fermilab alignment survey) | −11.565000 | 993.350283 | −3.643551 | mczero → minos-fd |
| NOvA FD upstream face centre (Fermilab alignment survey) | −11,037.455993 | 810,422.323775 | −4,162.644422 | mczero → minos-fd |

heights of the note are ellipsoidal, as the `$POLYP` altitude confirms), the MINOS FD supermodule midpoint [12] used to define the aiming line, and the 2014-note antenna-derived endpoints that are retained for comparison.

Table 3: Other input positions. FNAL antenna coordinates are quoted to 1″ (about 31 m N–S, 23 m E–W); Ash River coordinates to $10^{-6}$ arc-minute.

| id | site | position as given | source |
|---|---|---|---|
| target | NuMI target (TDR 1998) upstream face | FSCS $x, y, z$ = 100327.838, 97780.188, 626.743 ft | NuMI TDR 1998 App. A beam sheet (design); Beams-doc-1148 projection |
| mczero | MCZERO horn 1 upstream face, (TDR 1998) | FSCS $x, y, z$ = 100324.941, 97781.717, 626.552 ft | NuMI TDR 1998 App. A beam sheet row 0089 IHHRN1 |
| decay-end | NuMI decay pipe end (TDR 1998) | FSCS $x, y, z$ = 98230.009, 98887.039, 488.477 ft | NuMI TDR 1998 App. A beam sheet row 0092 DENCU |
| fnal-mast | FNAL MINOS Surface Building mast antenna | 41° 50’ 23.0000” N, 88° 16’ 13.0000” W, $h$ = 228.000 m (ellipsoid) | A.Norman 2014 note Table 2 (MINOS-based TDU receivers) |
| nd-under-mast | ND endpoint if it were under the MINOS mast (105 + 15 m down) | 41° 50’ 23.0000” N, 88° 16’ 13.0000” W, $h$ = 228.000 m (ellipsoid) | 2014 note sec. 3.1 and Tables 2-3 (superseded: the ND is not under the mast) |
| fd1 | Ash River Ant-1 (FD front-face centre) | 48° 22’ 42.9468” N, 92° 49’ 52.6158” W, $h$ = 348.000 m (ellipsoid) | 2014 note Table 2 |
| fd2 | Ash River Ant-2 fix, front face via -19.8 m (antenna-derived, superseded) | 48° 22’ 42.9468” N, 92° 49’ 52.6158” W, $h$ = 348.638 m (ellipsoid) | 2014 note Table 2 (antenna used by the 2013 notebook) |
| fd3 | Ash River Ant-3 (FD front-face centre) | 48° 22’ 42.9850” N, 92° 49’ 52.6583” W, $h$ = 350.346 m (ellipsoid) | 2014 note Table 2 / $GPGGA 2013-03-07 21:03:41 UTC |
| minos-fd | MINOS Far Detector (supermodule midpoint, Soudan) | 47° 49’ 12.9612” N, 92° 14’ 29.0832” W, $h$ = −248.400 m (ellipsoid) | MINOS detectors NIM A 596 (2008) 190, sec. 6.5 |
| vermilion | Lake Vermilion (100 ft below the surface) | 47° 50’ 49.4300” N, 92° 17’ 46.3400” W, $H$ = 383.440 m (MSL), $N_g$ = −30.892 m | Coordinates supplied by A. Norman (2026-11-09); surface 1,358 ft (Lake Vermilion, MN DNR / Vermilion Dam); GEOID18 |
| two-harbors | Lake Superior, Two Harbors (100 ft below the surface) | 46° 59’ 55.2000” N, 91° 38’ 16.6500” W, $H$ = 152.735 m (MSL), $N_g$ = −29.103 m | Coordinates supplied by A. Norman (2026-11-09); surface 601.1 ft IGLD85 (NOAA chart datum); GEOID18 |

### 3.4 Lake sites

Two additional points were added at 100 ft below the lake surface: Lake Vermilion at 47°50′49.43″ N, 92°17′46.34″ W and Lake Superior at Two Harbors, 46°59′55.2″ N, 91°38′16.65″ W. The lake surfaces are 1,358 ft (Vermilion Dam normal pool, NGVD29) and 601.1 ft (Lake Superior chart datum, IGLD 1985), so the points are at orthometric heights $H$ = 1,258 ft = 383.440 m and $H$ = 501.1 ft = 152.735 m. These are mean-sea-level heights and are converted to ellipsoidal heights with the GEOID18 separations at the two points, $N_g$ = −30.892 m and −29.103 m (NGS geoid service); the conversion is a ledger line (Table 20, Sec. 7). Lake levels vary by about ±0.6 m, worth 0.04 m of

chord (0.01 ticks), and the NGVD29/IGLD85 to NAVD88 datum differences are at the decimetre level.

### 3.5 Converted positions

Table 4 gives every endpoint used below in geodetic (WGS84) and ECEF coordinates after all conversions (site projection, beamline frame, antenna offsets); Table 5 repeats the principal points in decimal degrees.

Table 4: Endpoints in WGS84 geodetic and ECEF coordinates. `nd/fd` are the surveyed faces, `-left/-right` their edges; `nd-1015`, `nd-under-mast` and `fd2` are superseded geometries (Appendix A).

| id | latitude | longitude | $h$ (m) | $X$ (m) | $Y$ (m) | $Z$ (m) |
|---|---|---|---|---|---|---|
| target | 41° 49’ 55.0917” N | 88° 15’ 55.9086” W | 158.074 | 144,061.137 | −4,757,403.941 | 4,231,821.378 |
| mczero | 41° 49’ 55.1213” N | 88° 15’ 55.9261” W | 158.016 | 144,060.713 | −4,757,403.302 | 4,231,822.020 |
| decay-end | 41° 50’ 16.5133” N | 88° 16’ 08.6061” W | 115.930 | 143,754.011 | −4,756,940.784 | 4,232,285.708 |
| nd | 41° 50’ 24.3467” N | 88° 16’ 13.7981” W | 96.572 | 143,628.970 | −4,756,768.845 | 4,232,452.856 |
| nd-left | 41° 50’ 24.3203” N | 88° 16’ 13.8777” W | 96.572 | 143,627.151 | −4,756,769.443 | 4,232,452.250 |
| nd-right | 41° 50’ 24.3731” N | 88° 16’ 13.7185” W | 96.572 | 143,630.788 | −4,756,768.247 | 4,232,453.463 |
| fd | 48° 22’ 41.5413” N | 92° 49’ 50.7737” W | 337.611 | −209,624.380 | −4,239,417.585 | 4,745,165.129 |
| fd-left | 48° 22’ 41.4272” N | 92° 49’ 51.1078” W | 337.623 | −209,631.378 | −4,239,419.886 | 4,745,162.796 |
| fd-right | 48° 22’ 41.6555” N | 92° 49’ 50.4396” W | 337.599 | −209,617.382 | −4,239,415.284 | 4,745,167.462 |
| minos-fd | 47° 49’ 12.9612” N | 92° 14’ 29.0832” W | −248.400 | −167,796.989 | −4,287,098.683 | 4,703,296.906 |
| vermilion | 47° 50’ 49.4300” N | 92° 17’ 46.3400” W | 352.548 | −171,824.466 | −4,285,132.713 | 4,705,742.418 |
| two-harbors | 46° 59’ 55.2000” N | 91° 38’ 16.6500” W | 123.632 | −124,565.097 | −4,356,099.794 | 4,641,754.113 |
| nd-1015 | 41° 50’ 24.9725” N | 88° 16’ 14.1694” W | 104.124 | 143,620.189 | −4,756,761.853 | 4,232,472.279 |
| nd-under-mast | 41° 50’ 23.0000” N | 88° 16’ 13.0000” W | 108.000 | 143,648.469 | −4,756,804.502 | 4,232,429.524 |
| fd2 | 48° 22’ 42.9468” N | 92° 49’ 52.6158” W | 328.838 | −209,660.350 | −4,239,377.478 | 4,745,187.407 |

Table 5: Principal points in decimal degrees (WGS84).

| id | latitude (°) | longitude (°, east +) | $h$ (m) |
|---|---|---|---|
| target | 41.831969923 | −88.265530166 | 158.074 |
| mczero | 41.831978142 | −88.265535036 | 158.016 |
| nd | 41.840096298 | −88.270499477 | 96.572 |
| fd | 48.378205927 | −92.830770475 | 337.611 |
| minos-fd | 47.820267000 | −92.241412000 | −248.400 |
| vermilion | 47.847063889 | −92.296205556 | 352.548 |
| two-harbors | 46.998666667 | −91.637958333 | 123.632 |

## 4 Distances

### 4.1 Chords in the spherical and ellipsoidal models

Neutrinos travel the Euclidean chord $L = |\vec{p}_B - \vec{p}_A|$ through the Earth, not the surface arc. Table 6 gives, for every baseline, the chord in the spherical reference model, in the WGS84 model with the surveyed or given points, and after the antenna-to-endpoint offsets where any apply.

Table 6: Chord length by Earth model. $L_{\mathrm{sph}}$: sphere of radius $a$; $L_{\mathrm{WGS}}$: ellipsoid, points as given; $L_{\mathrm{final}}$: after antenna offsets (none for surveyed faces).

| baseline | $L_{\mathrm{sph}}$ (m) | $L_{\mathrm{WGS}}$ (m) | $\Delta$ (m) | $L_{\mathrm{final}}$ (m) | $\Delta$ (m) |
|---|---|---|---|---|---|
| ND upstream face → FD upstream face (both surveyed) | 810,337.734 | 809,514.750 | −822.984 | 809,514.750 | 0.000 |
| NuMI target → FD upstream face (surveyed) | 811,333.740 | 810,509.171 | −824.569 | 810,509.171 | 0.000 |
| NuMI target → ND upstream face (surveyed) | 996.010 | 994.424 | −1.585 | 994.424 | 0.000 |
| MCZERO → FD upstream face (surveyed; stationing 810,422.324 m) | 811,332.738 | 810,508.171 | −824.567 | 810,508.171 | 0.000 |
| MCZERO → ND upstream face (surveyed; stationing 993.350 m) | 995.008 | 993.424 | −1.584 | 993.424 | 0.000 |
| ND at the quoted 1015 m → FD (superseded ND placement) | 810,317.097 | 809,494.148 | −822.949 | 809,494.148 | 0.000 |
| ND under the mast → FD from the Ant-2 fix (superseded 2014-note geometry) | 810,448.173 | 809,625.125 | −823.048 | 809,616.263 | −8.862 |
| NuMI target → Lake Vermilion (-100 ft) | 740,693.015 | 739,883.535 | −809.480 | 739,881.733 | −1.802 |
| ND upstream face → Lake Vermilion (-100 ft) | 739,697.049 | 738,889.155 | −807.894 | 738,887.353 | −1.802 |
| NuMI target → Lake Superior, Two Harbors (-100 ft) | 634,190.629 | 633,444.355 | −746.274 | 633,442.910 | −1.446 |
| ND upstream face → Lake Superior, Two Harbors (-100 ft) | 633,194.760 | 632,450.073 | −744.687 | 632,448.627 | −1.446 |
| NuMI target → MINOS FD (validation of the target position) | 736,157.346 | 735,344.614 | −812.732 | 735,344.614 | 0.000 |
| NuMI target → decay-pipe end (TDR design axis) | 725.363 | 724.184 | −1.179 | 724.184 | 0.000 |

## 4.2 Correction ledger

The ledger lists each change of the endpoints in order with its effect on $L$, $t$ and the tick count. For the surveyed faces only the Earth model enters (Table 7); the 2014-note geometry with its antenna offsets is given for comparison in Table 26 of Appendix A.

Table 7: Correction ledger, ND face → FD face.

| step | applied | $L$ (m) | $\Delta L$ (m) | $\Delta t$ (ns) | $\Delta$ticks |
|---|---|---|---|---|---|
| reference model | spherical Earth, R = 6378137.000 m; heights as given; surveyed points, no offsets | 810,337.734 | 0.000 | 0.00 | 0.000 |
| Earth model | WGS84 ellipsoid; heights as given; surveyed points, no offsets | 809,514.750 | −822.984 | −2,745.18 | −175.691 |

## 4.3 Chord geometry

The geocentric central angle $\gamma = \angle(\vec{p}_A, \vec{p}_B)$, the surface arc at the mean geocentric radius $\bar{R}\gamma$, the sagitta $\bar{R}(1 - \cos\gamma/2)$, and the azimuth and elevation of the chord in the local frames of Eq. (6) at each end are given in Table 8. A great-circle distance would be wrong by more than 100 ticks.

Table 8: Chord geometry. Elevation is negative when the chord dips below the local horizontal at the source; at the target end a positive value means the chord arrives rising.

| chord | $\gamma$ (°) | arc (m) | arc−chord (m) | sagitta (m) | azimuth (°) | dip at source (°) | elev. at target (°) | deepest $h$ (m) |
|---|---|---|---|---|---|---|---|---|
| ND → FD | 7.28884 | 810,057.9 | 543.16 | 12,877.0 | 335.30891 | −3.62569 | 3.65835 | −12,651 |
| target → FD | 7.29777 | 811,054.2 | 545.00 | 12,908.6 | 335.31237 | −3.63452 | 3.65845 | −12,652 |
| target → ND | 0.00893 | 992.3 | −2.09 | 0.0 | 335.42658 | −3.55027 | −3.54134 | 97 |
| target → MINOS FD | 6.62038 | 735,749.1 | 404.52 | 10,623.8 | 336.09421 | −3.34056 | 3.27617 | −10,664 |
| target → Lake Vermilion | 6.66098 | 740,295.7 | 414.00 | 10,755.0 | 335.91399 | −3.31416 | 3.34306 | −10,494 |
| target → Two Harbors | 5.70183 | 633,701.3 | 258.44 | 7,881.3 | 336.09597 | −2.85316 | 2.84608 | −7,737 |

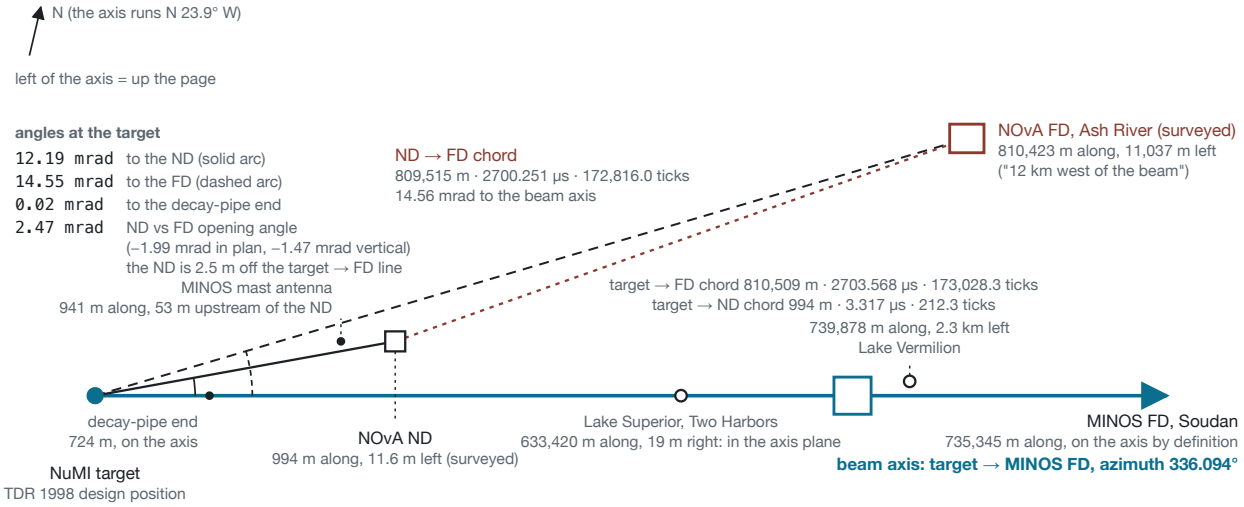



Figure 1: Plan view, not to scale: the beam axis from the NuMI target through the MINOS far detector, the surveyed ND and FD faces to its left, the MINOS mast antenna, and the chords used in the calculation.

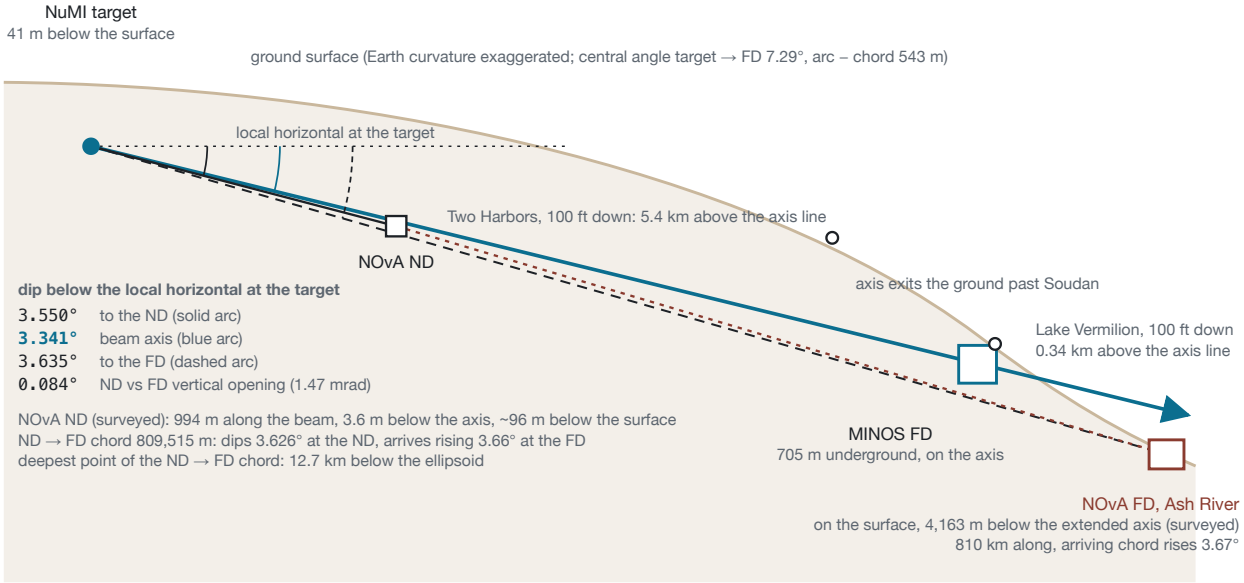



Figure 2: Elevation view, not to scale: dips of the chords below the target's local horizontal, the curved ground surface, and the positions of the detectors relative to the extended beam axis.

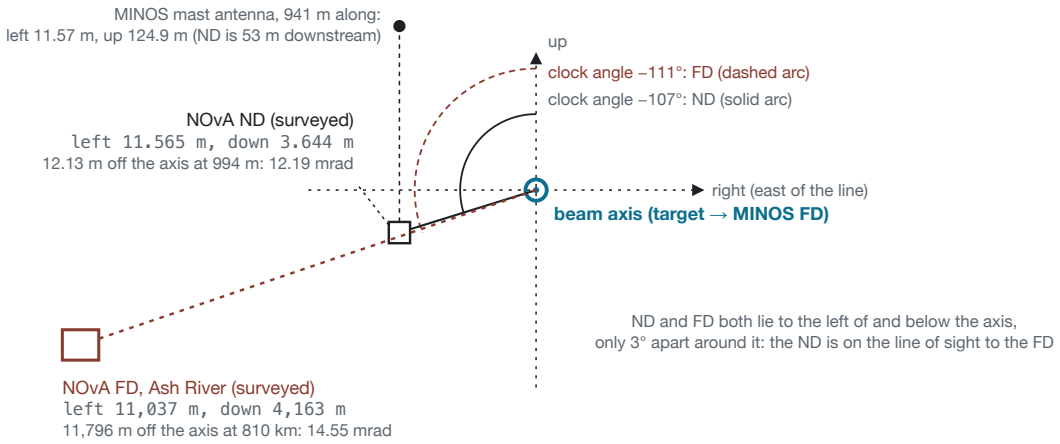



Figure 3: Transverse view looking downstream along the beam axis: the surveyed ND and FD faces lie to the left of and below the axis at almost the same clock angle.

# 5 Time of flight and timing offsets

## 5.1 Flight time and clock ticks

With $c = 299{,}792{,}458\,\mathrm{m/s}$ exactly and neutrinos taken to travel at $c$ (the mass-induced delay is evaluated in Sec. 5.4),

$$t = \frac{L}{c}, \qquad n = f_{\mathrm{clk}}\, t, \qquad f_{\mathrm{clk}} = 64\,\mathrm{MHz}, \tag{12}$$

so one tick is 15.625 ns or 4.684 m of light path. The legacy Global Trigger unit of 500 ns is also given. Table 9 lists the results for the surveyed geometry; the integer tick count uses round-to-nearest and the residual is the difference between the integer and the exact value.

Table 9: Flight times for the surveyed geometry. "total" includes the Main Injector shift of Sec. 5.2.

| chord | $L$ (m) | $t$ (µs) | ticks | integer | residual (ns) | 500 ns units | total ticks |
|---|---|---|---|---|---|---|---|
| ND upstream face → FD upstream face (both surveyed) | 809,514.750 | 2,700.250552 | 172,816.035 | 172,816 | −0.55 | 5,400.5011 | 272,736.417 |
| NuMI target → FD upstream face (surveyed) | 810,509.171 | 2,703.567584 | 173,028.325 | 173,028 | −5.08 | 5,407.1352 | 272,948.707 |
| NuMI target → ND upstream face (surveyed) | 994.424 | 3.317043 | 212.291 | 212 | −4.54 | 6.6341 | 100,132.672 |
| MCZERO → FD upstream face (surveyed; stationing 810,422.324 m) | 810,508.171 | 2,703.564248 | 173,028.112 | 173,028 | −1.75 | 5,407.1285 | 272,948.494 |
| MCZERO → ND upstream face (surveyed; stationing 993.350 m) | 993.424 | 3.313707 | 212.077 | 212 | −1.21 | 6.6274 | 100,132.459 |
| NuMI target → Lake Vermilion (-100 ft) | 739,881.733 | 2,467.979809 | 157,950.708 | 157,951 | 4.57 | 4,935.9596 | 257,871.089 |
| ND upstream face → Lake Vermilion (-100 ft) | 738,887.353 | 2,464.662914 | 157,738.426 | 157,738 | −6.66 | 4,929.3258 | 257,658.808 |
| NuMI target → Lake Superior, Two Harbors (-100 ft) | 633,442.910 | 2,112.938110 | 135,228.039 | 135,228 | −0.61 | 4,225.8762 | 235,148.421 |
| ND upstream face → Lake Superior, Two Harbors (-100 ft) | 632,448.627 | 2,109.621539 | 135,015.779 | 135,016 | 3.46 | 4,219.2431 | 234,936.160 |

### 5.2 Main Injector timing shift

After the 2012 shutdown the beam moved relative to the Main Injector Beam Sync event MIBS \$74 by 141 MI revolutions [3]. With $h_{\mathrm{MI}} = 588$ RF buckets per turn at the flat-top RF frequency $f_{\mathrm{RF}}$,

$$T_{\mathrm{rev}} = \frac{h_{\mathrm{MI}}}{f_{\mathrm{RF}}}, \qquad \Delta t_{\mathrm{MI}} = 141\, T_{\mathrm{rev}}, \qquad n_{\mathrm{MI}} = f_{\mathrm{clk}}\, \Delta t_{\mathrm{MI}}, \tag{13}$$

giving Table 10. One revolution is 708.655 ticks: the 53.1034 MHz RF and the 64 MHz clock are not commensurate, and a 1 kHz error in $f_{\mathrm{RF}}$ shifts $n_{\mathrm{MI}}$ by 1.882 ticks.

Table 10: Main Injector 141-turn shift.

| turns | buckets/turn | $f_{\mathrm{RF}}$ (MHz) | $T_{\mathrm{rev}}$ (µs) | ticks/turn | $\Delta t_{\mathrm{MI}}$ (µs) | $n_{\mathrm{MI}}$ (ticks) |
|---|---|---|---|---|---|---|
| 141 | 588 | 53.1034 | 11.072737 | 708.655 | 1,561.255965 | 99,920.382 |

### 5.3 Combined offset and trigger window

The total offset of the FD trigger relative to MIBS \$74 is $n_{\mathrm{tot}} = n_{\mathrm{ToF}} + n_{\mathrm{MI}}$ (Table 11); the Sagnac term is reported but not added. The window of Eq. (1) in ticks is given in Table 12; a 50 µs window spans 3,200 ticks and overlaps one or two 50 µs microslices, so a recorded beam event is up to 100 µs wide.

Table 11: Combined offset, ND face → FD face plus the MI shift.

| $n_{\mathrm{ToF}}$ | $n_{\mathrm{MI}}$ | $n_{\mathrm{tot}}$ | integer | residual (ns) |
|---|---|---|---|---|
| 172,816.035 | 99,920.382 | 272,736.417 | 272,736 | −6.52 |

Table 12: Trigger window $[t - 20\,\mu\mathrm{s},\ t + 30\,\mu\mathrm{s})$ in ticks after $t_0$, relative to the flight time alone and to the flight time plus the MI shift.

| ToF low | ToF high | total low | total high | ticks/microslice | microslices overlapped |
|---|---|---|---|---|---|
| 171,536.04 | 174,736.04 | 271,456.42 | 274,656.42 | 3,200 | 1–2 |

### 5.4 Effects outside the geometric flight time

In the rotating ECEF frame a signal from $A$ to $B$ acquires the Sagnac correction $\Delta t_{\mathrm{S}} = \omega_{\oplus}\,(x_A y_B - y_A x_B)/c^2$ with $\omega_{\oplus} = 7.2921150 \times 10^{-5}$ rad/s [13]; it is negative here because the beam travels westward and the FD rotates toward it. A neutrino of mass $m$ and energy $E$ is delayed by $\delta t = t\, m^2/(2E^2)$. Antenna cable delays ($\ell/(v_f c)$) are receiver calibrations and not part of the flight time. Table 13 lists them; the MINOS neutrino-speed measurement [14] treats the same terms.

Table 13: Non-geometric terms for the ND → FD chord.

| effect | ns | ticks | status |
|---|---|---|---|
| Sagnac (Earth rotation, ECEF frame) | −1.303 | −0.0834 | reported, not applied |
| neutrino mass delay ($m = 0.1\,\mathrm{eV}$, $E = 2\,\mathrm{GeV}$) | $3.38 \times 10^{-15}$ | $2.16 \times 10^{-16}$ | negligible |
| antenna cable delay, NOvA ND upstream face centre (Fermilab alignment survey) | 119.61 | 7.655 | receiver calibration, not ToF |
| antenna cable delay, NOvA FD upstream face centre (Fermilab alignment survey) | 119.61 | 7.655 | receiver calibration, not ToF |

# 6 Angles relative to the beamline

## 6.1 Beam axis and off-axis positions

The beam axis of this note is the line from the target through the MINOS FD midpoint (Table 14); the TDR design axis agrees with it in direction to 0.02 mrad. Table 15 gives every site's off-axis angle at the target, distance along the axis and transverse position from Eq. (11).

Table 14: The beam axis.

| axis | length (m) | azimuth at origin (°) | elevation at origin (°) | unit vector (ECEF) |
|---|---|---|---|---|
| NuMI beam axis (target → MINOS FD) | 735,344.614 | 336.09421 | −3.34056 | (-0.424097927, 0.639571229, 0.641162687) |

Table 15: Sites relative to the beam axis: off-axis angle at the target, distance along the axis, transverse distance with right/up components, and clock angle (0° above, 90° right).

| id | off-axis (mrad) | along (m) | transverse (m) | right (m) | up (m) | clock (°) |
|---|---|---|---|---|---|---|
| target | 0.0000 | 0.000 | 0.000 | 0.000 | 0.000 | – |
| mczero | 0.1459 | 1.000 | 0.000 | 0.000 | 0.000 | 64.88 |
| decay-end | 0.0197 | 724.184 | 0.014 | 0.000 | 0.014 | −1.89 |
| nd | 12.1935 | 994.350 | 12.125 | −11.565 | −3.643 | −107.49 |
| fd | 14.5547 | 810,423.324 | 11,796.315 | −11,037.456 | −4,162.644 | −110.66 |
| minos-fd | 0.0000 | 735,344.614 | 0.000 | 0.000 | 0.000 | – |
| vermilion | 3.1739 | 739,878.006 | 2,348.293 | −2,323.438 | 340.763 | −81.66 |
| two-harbors | 8.5068 | 633,419.990 | 5,388.531 | 19.357 | 5,388.496 | 0.21 |
| nd-1015 | 12.4658 | 1,015.000 | 12.653 | −11.570 | 5.123 | −66.12 |
| nd-under-mast | 13.3426 | 948.316 | 12.654 | −11.570 | 5.123 | −66.12 |
| fd2 | 14.5753 | 810,478.514 | 11,813.783 | −11,051.360 | −4,175.273 | −110.70 |
| fnal-mast | 132.4921 | 941.341 | 125.455 | −11.570 | 124.921 | −5.29 |

The lake sites are discussed in Sec. 7.

## 6.2 Opening angles

Table 16 gives the opening angles at the target between pairs of chords, their horizontal and vertical components, the transverse distance of the first endpoint from the second chord, and each chord's

angle to the beam axis. With both surveyed faces the ND lies only 2.46 m off the target → FD line: the Near Detector sits on the line of sight to the Far Detector.

Table 16: Opening angles at the NuMI target.

| angle | total (mrad) | horizontal (mrad) | vertical (mrad) | $a$ transverse to $Vb$ (m) | $a$ vs axis (mrad) | $b$ vs axis (mrad) |
|---|---|---|---|---|---|---|
| Target: ND vs FD (both surveyed) | 2.4740 | −1.9895 | −1.4705 | 2.460 | 12.1935 | 14.5547 |
| Target: ND (surveyed) vs MINOS FD | 12.1935 | 11.6313 | 3.6600 | 12.125 | 12.1935 | 0.0000 |
| Target: ND at the quoted 1015 m vs surveyed ND | 8.7139 | −0.2374 | −8.7107 | 8.845 | 12.4658 | 12.1935 |
| Target: ND under the mast (2014 note) vs surveyed ND | 9.0840 | 0.5638 | −9.0665 | 8.615 | 13.3426 | 12.1935 |
| Target: FD from the Ant-2 fix vs surveyed FD | 0.0223 | 0.0162 | 0.0152 | 18.037 | 14.5753 | 14.5547 |
| Target: ND vs Lake Vermilion | 9.4386 | 8.4915 | 4.1208 | 9.386 | 12.1935 | 3.1739 |
| Target: ND vs Lake Superior (Two Harbors) | 16.8551 | 11.6647 | 12.1668 | 16.760 | 12.1935 | 8.5068 |

### 6.3 Near Detector face edges

The ND face is 158.125 in square (half-width 2.0081875 m). Table 17 gives the positions of its centre and edges and their angles to the beam axis; the face subtends 4.0386 mrad (left–right) and 4.0389 mrad (top–bottom) at the target. Table 18 gives the opening angle between each face point and the target → FD chord. Figure 4 shows the face with the iso-angle contours. The FD face (half-size 304.21 in) is treated identically; its edges are 0.019 mrad apart as seen from the target.

Table 17: ND face points: geodetic position, angle to the beam axis and transverse offsets from the axis.

| point | latitude | longitude | $h$ (m) | angle to axis (mrad) | right (m) | up (m) | clock (°) |
|---|---|---|---|---|---|---|---|
| left | 41° 50’ 24.3203” N | 88° 16’ 13.8777” W | 96.572 | 14.1325 | −13.573 | −3.643 | −105.03 |
| center | 41° 50’ 24.3467” N | 88° 16’ 13.7981” W | 96.572 | 12.1935 | −11.565 | −3.643 | −107.49 |
| right | 41° 50’ 24.3731” N | 88° 16’ 13.7185” W | 96.572 | 10.2854 | −9.557 | −3.643 | −110.87 |
| top | 41° 50’ 24.3501” N | 88° 16’ 13.8002” W | 98.577 | 11.7457 | −11.565 | −1.635 | −98.05 |
| bottom | 41° 50’ 24.3432” N | 88° 16’ 13.7961” W | 94.568 | 12.9444 | −11.565 | −5.652 | −116.04 |

Table 18: Opening angle at the target between each ND face point and the chord to the FD face.

| point | opening angle (mrad) | horizontal (mrad) | vertical (mrad) | transverse to the FD chord (m) | chord vs axis (mrad) |
|---|---|---|---|---|---|
| left | 1.4724 | 0.0296 | −1.4721 | 1.464 | 14.1325 |
| center | 2.4740 | −1.9895 | −1.4705 | 2.460 | 12.1935 |
| right | 4.2695 | −4.0088 | −1.4691 | 4.246 | 10.2854 |
| top | 4.0179 | −1.9910 | −3.4899 | 3.995 | 11.7457 |
| bottom | 2.0624 | −1.9880 | 0.5489 | 2.051 | 12.9444 |

## 7 Lake sites: Lake Vermilion and Lake Superior at Two Harbors

The two lake points of Sec. 3.4 are carried through the same chain as the detectors. Table 19 summarises the chords from the target and from the ND, the flight times and the positions relative to the beam axis; Table 20 gives the correction ledgers, in which the mean-sea-level to ellipsoid conversion appears as its own step (lowering the far end by $|N_g|$ shortens the chord by $|N_g|\sin(\text{dip})$, 1.80 m and 1.45 m). Figures 5–7 show the geometry.

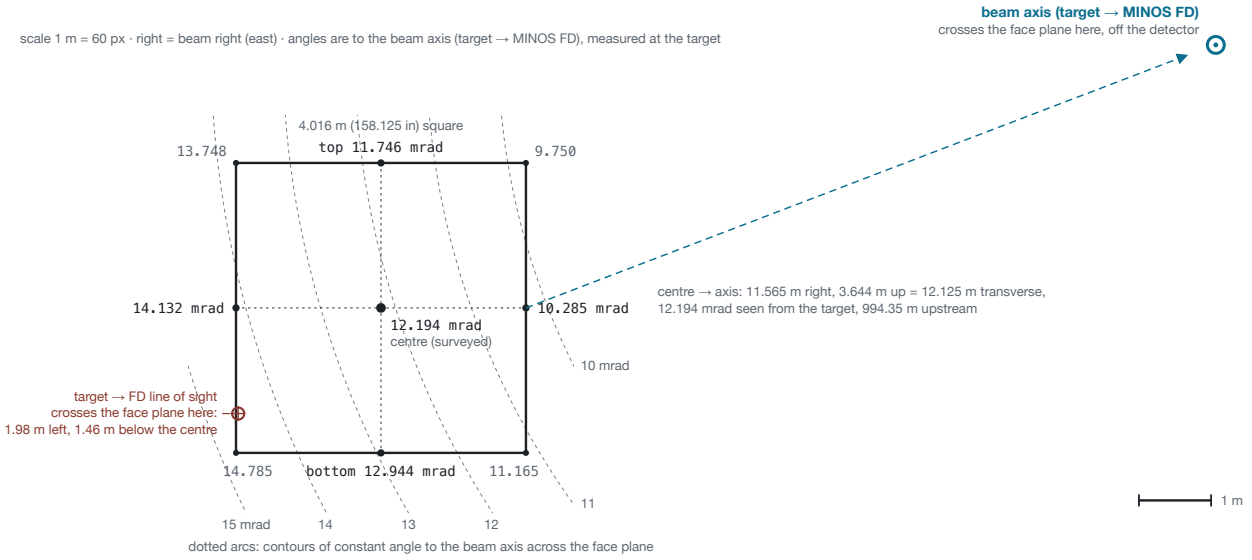



Figure 4: The ND front face looking downstream, to scale: the angle to the beam axis at the centre, edges and corners, contours of constant angle across the face plane, the point where the beam axis crosses the face plane (off the detector, to the right and above), and the point where the target → FD line of sight crosses the face plane.

Lake Vermilion lies 740 km along the axis, 5 km past Soudan, 2.3 km to the left of the axis and 0.34 km above its extension, so the chord from the target makes 3.17 mrad with the axis, almost all of it in plan. The Two Harbors point lies almost exactly in the vertical plane of the axis (19 m to the right) 633 km along it; because the ground curves away from the straight beam and the axis climbs back toward the surface only past Soudan, the point is 5.4 km above the extended axis line and the chord from the target makes 8.51 mrad with the axis, almost all of it vertical (clock angle 0.2°: directly above the beam). The opening angles at the target between the ND chord and the lake chords are 9.44 mrad (Vermilion) and 16.86 mrad (Two Harbors).

Table 19: Lake sites: chords and flight times from the target and from the ND face, and position relative to the beam axis.

| site | $L$ from target (m) | $t$ (µs) | ticks | $L$ from ND (m) | ticks | angle to axis (mrad) | along (m) | right (m) | up (m) |
|---|---|---|---|---|---|---|---|---|---|
| Lake Vermilion | 739,881.733 | 2,467.979809 | 157,950.708 | 738,887.353 | 157,738.426 | 3.1739 | 739,878.0 | −2,323.4 | 340.8 |
| Lake Superior, Two Harbors | 633,442.910 | 2,112.938110 | 135,228.039 | 632,448.627 | 135,015.779 | 8.5068 | 633,420.0 | 19.4 | 5,388.5 |

# 8 Validation

The target position and the realisation of the beamline frame rest on design documents; Table 21 checks them against independently published or surveyed quantities. The target → MINOS FD chord agrees with the published 735,338.0 m (from the horn-1 upstream end) to 7 m, the beam azimuth with the 1999 geodetic survey to 13″, the FD ellipsoidal height with the NAVD88 survey elevation to 0.3 m when the GEOID18 separation at Ash River (−31.221 m) is used (1.5 m with the receiver's own −32.462 m), and the MCZERO → FD chord equals $\sqrt{X^2+Y^2+Z^2}$ of the survey coordinates identically. The surveyed ND confirms the 2014 antenna's 11.57 m left offset to 5 mm. In addition the 2013 notebook is reproduced to $10^{-9}$ (Appendix A).

Table 20: Correction ledgers of the chords from the NuMI target to the two lake sites.

| site | step | $L$ (m) | $\Delta L$ (m) | $\Delta t$ (ns) | $\Delta$ticks |
|---|---|---|---|---|---|
| Lake Vermilion (-100 ft) | reference model | 740,693.015 | 0.000 | 0.00 | 0.000 |
| Lake Vermilion (-100 ft) | Earth model | 739,883.535 | −809.480 | −2,700.13 | −172.809 |
| Lake Vermilion (-100 ft) | height datum | 739,881.733 | −1.802 | −6.01 | −0.385 |
| Lake Superior, Two Harbors (-100 ft) | reference model | 634,190.629 | 0.000 | 0.00 | 0.000 |
| Lake Superior, Two Harbors (-100 ft) | Earth model | 633,444.355 | −746.274 | −2,489.30 | −159.315 |
| Lake Superior, Two Harbors (-100 ft) | height datum | 633,442.910 | −1.446 | −4.82 | −0.309 |

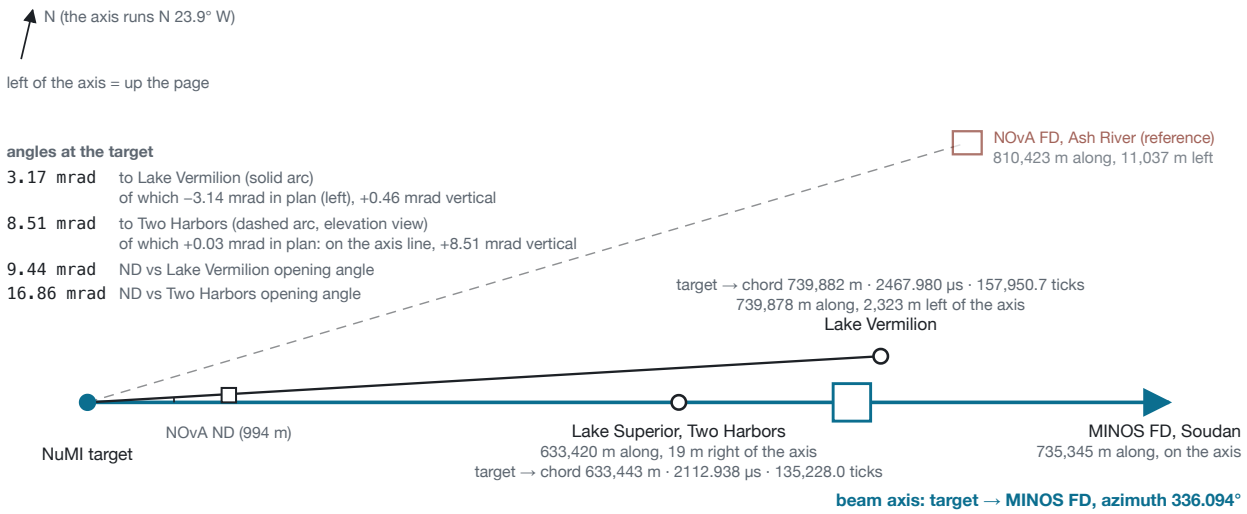



Figure 5: Lake sites, plan view, not to scale: the Two Harbors point in the vertical plane of the beam axis at 633 km, Lake Vermilion 740 km along and 2.3 km to the left; the MINOS and Ash River far detectors for reference.

Table 21: Validation against published and surveyed quantities.

| quantity | this note | reference | difference |
|---|---|---|---|
| target → MINOS FD midpoint | 735,344.6 m | 735,338.0 m from horn 1 [12] | 6.6 m |
| beam azimuth at the target | 336.09421° | 336.09788° [10] | −0.00367° |
| beam dip at the target | 3.34056° | 3.34349° [15] | −0.00293° |
| decay-pipe length (TDR stations) | 675.0 m | 675 m [15] | – |
| FD ellipsoidal height from the survey frame | 337.611 m | $H + N_g = 368.535 - 31.221 =$ 337.314 m (GEOID18) | 0.30 m |
| same, with the receiver's geoid separation | 337.611 m | $368.535 - 32.462 = 336.073$ m | 1.54 m |
| MCZERO → FD chord vs $\lvert(X, Y, Z)\rvert$ | 810,508.1714 m | 810,508.1714 m | 0.000000 m |
| off-axis angle of the FD face | 14.555 mrad | 14.6 mrad [16] | – |

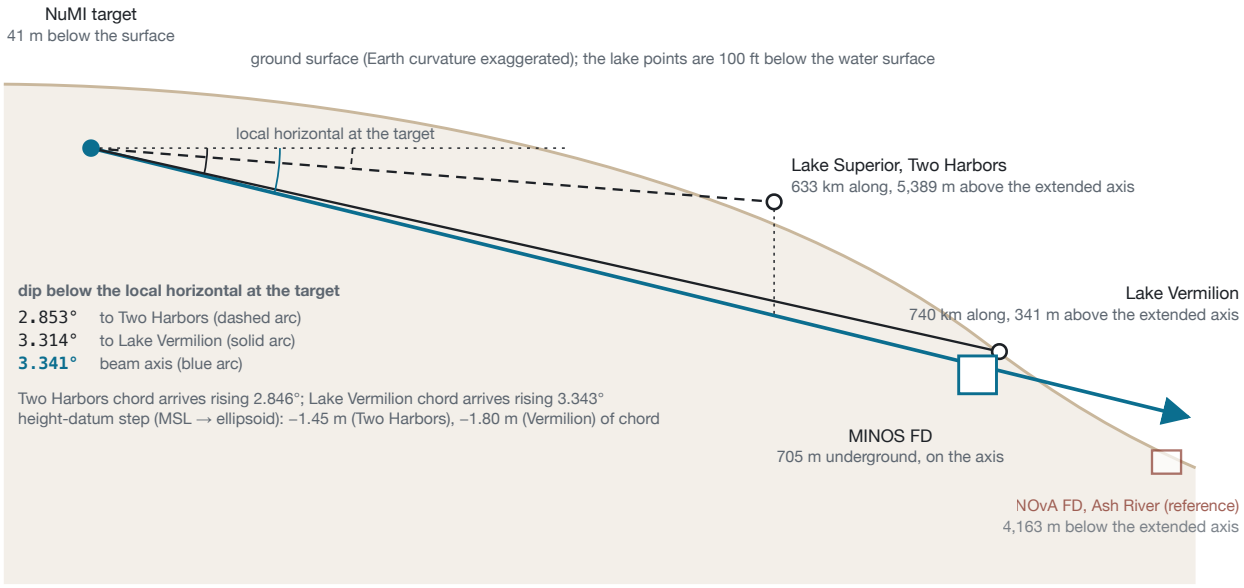



Figure 6: Lake sites, elevation view, not to scale: both points lie just below the ground surface above the extended beam axis, Two Harbors 5.4 km and Lake Vermilion 0.34 km above it; the chords from the target dip 2.853° and 3.314° against 3.341° for the axis.

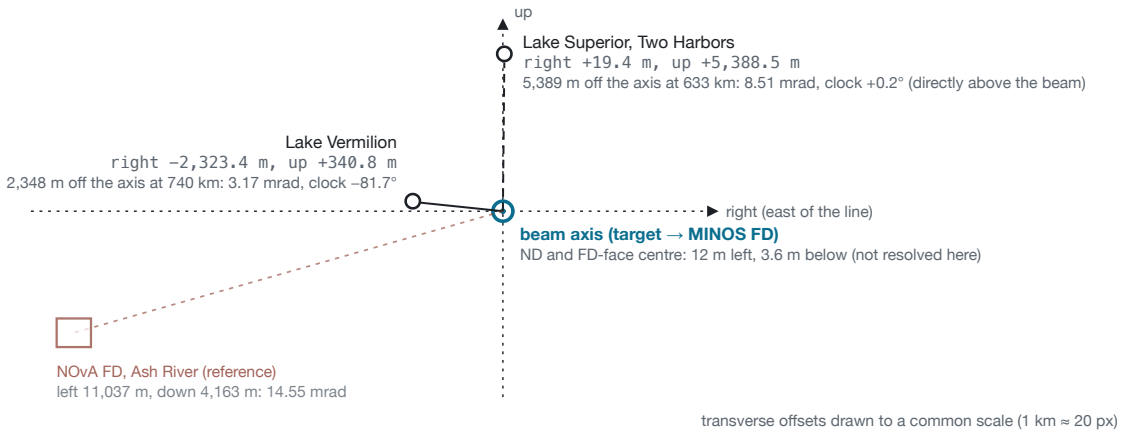



Figure 7: Lake sites, transverse view looking downstream along the beam axis, offsets to a common scale: Two Harbors directly above the axis at 5.4 km, Lake Vermilion 2.3 km to the left and slightly above, the Ash River face to the lower left for reference.

## 9 Sensitivities and error budget

Numerical partial derivatives of the ND → FD chord with respect to each input are given in Table 22; the gradient of $L$ with respect to an endpoint position is the unit chord vector, so an endpoint error of $\sigma$ metres along the chord is worth $\sigma$ metres of $L$ (3.34 ns/m). The error budget (Table 23) combines the declared $1\sigma$ uncertainties in quadrature and lists conventions and neglected terms separately. With both faces surveyed, the budget is dominated by the RF-frequency term of the Main Injector shift; combined items: root-sum-square 1.906 ticks (29.8 ns), linear sum 2.309 ticks (36.1 ns) .

Table 22: Sensitivities of the ND → FD chord.

| variation | $\Delta L$ (m) | $\Delta t$ (ns) | $\Delta$ticks |
|---|---|---|---|
| NOvA ND upstream face centre (Fermilab alignment survey): height +1 m | 0.0632 | 0.211 | 0.0135 |
| NOvA ND upstream face centre (Fermilab alignment survey): latitude +1” | −27.9763 | −93.319 | −5.9724 |
| NOvA ND upstream face centre (Fermilab alignment survey): longitude +1” | 9.6187 | 32.085 | 2.0534 |
| NOvA ND upstream face centre (Fermilab alignment survey): endpoint +1 m East | 0.4169 | 1.391 | 0.0890 |
| NOvA ND upstream face centre (Fermilab alignment survey): endpoint +1 m North | −0.9068 | −3.025 | −0.1936 |
| NOvA FD upstream face centre (Fermilab alignment survey): height +1 m | 0.0638 | 0.213 | 0.0136 |
| NOvA FD upstream face centre (Fermilab alignment survey): latitude +1” | 27.2369 | 90.852 | 5.8146 |
| NOvA FD upstream face centre (Fermilab alignment survey): longitude +1” | −9.6182 | −32.083 | −2.0533 |
| NOvA FD upstream face centre (Fermilab alignment survey): endpoint +1 m East | −0.4674 | −1.559 | −0.0998 |
| NOvA FD upstream face centre (Fermilab alignment survey): endpoint +1 m North | 0.8817 | 2.941 | 0.1882 |
| spherical Earth (R = 6378137 m) at the final endpoints | 822.9838 | 2,745.178 | 175.6914 |
| GRS80 ellipsoid instead of WGS84 | 0.0000 | 0.000 | 0.0000 |

## 10 Summary

Table 24 collects all baselines. For the surveyed ND and FD faces the flight path is 809,514.750 m, the flight time 2,700.250552 µs = 172,816.035 ticks, and the total offset with the 141-turn Main Injector shift 272,736.417 ticks, i.e. 272,736 ticks when rounded (residual −6.5 ns). From the target the FD face is 810,509.171 m (173,028.325 ticks) and the ND face 994.424 m (212.291 ticks). The ND and FD are 12.19 and 14.55 mrad off the Soudan axis, on the same side (clock −107° and −111°), and the ND face spans 4.04 mrad at the target. Chords from the target to the lake points are 739,881.733 m (Lake Vermilion, 157,950.708 ticks) and 633,442.910 m (Two Harbors, 135,228.039 ticks). Open items are the as-built MCZERO and target positions (about 1 m along the beam) and the lake-level datums.

Table 23: Error budget, ND face → FD face, in 64 MHz ticks.

| source | ns | ticks | type | combined |
|---|---|---|---|---|
| NOvA ND upstream face centre (Fermilab alignment survey): endpoint offset ±1 m, worst case along the chord | 3.34 | 0.213 | systematic | yes |
| NOvA FD upstream face centre (Fermilab alignment survey): endpoint offset ±1 m, worst case along the chord | 3.34 | 0.213 | systematic | yes |
| Main Injector 141-turn shift (2012 shutdown): RF frequency ±1000 Hz | 29.40 | 1.882 | systematic | yes |
| integer rounding of the final tick count | 7.81 | 0.500 | convention | no |
| Sagnac term (not applied) | 1.30 | 0.083 | neglected | no |

Table 24: All baselines of the configuration. "total" adds the MI shift; the last column is the opening angle between the chord and the beam axis.

| baseline | $L$ (m) | $t$ (µs) | ticks | total ticks | integer | chord vs axis (mrad) |
|---|---|---|---|---|---|---|
| ND upstream face → FD upstream face (both surveyed) | 809,514.750 | 2,700.250552 | 172,816.035 | 272,736.417 | 272,736 | 14.5576 |
| NuMI target → FD upstream face (surveyed) | 810,509.171 | 2,703.567584 | 173,028.325 | 272,948.707 | 272,949 | 14.5547 |
| NuMI target → ND upstream face (surveyed) | 994.424 | 3.317043 | 212.291 | 100,132.672 | 100,133 | 12.1935 |
| MCZERO → FD upstream face (surveyed; stationing 810,422.324 m) | 810,508.171 | 2,703.564248 | 173,028.112 | 272,948.494 | 272,948 | 14.5547 |
| MCZERO → ND upstream face (surveyed; stationing 993.350 m) | 993.424 | 3.313707 | 212.077 | 100,132.459 | 100,132 | 12.2059 |
| ND at the quoted 1015 m → FD (superseded ND placement) | 809,494.148 | 2,700.181830 | 172,811.637 | 272,732.019 | 272,732 | 14.5618 |
| ND under the mast → FD from the Ant-2 fix (superseded 2014-note geometry) | 809,616.263 | 2,700.589164 | 172,837.706 | 272,758.088 | 272,758 | 14.5812 |
| NuMI target → Lake Vermilion (-100 ft) | 739,881.733 | 2,467.979809 | 157,950.708 | 257,871.089 | 257,871 | 3.1739 |
| ND upstream face → Lake Vermilion (-100 ft) | 738,887.353 | 2,464.662914 | 157,738.426 | 257,658.808 | 257,659 | 3.1634 |
| NuMI target → Lake Superior, Two Harbors (-100 ft) | 633,442.910 | 2,112.938110 | 135,228.039 | 235,148.421 | 235,148 | 8.5068 |
| ND upstream face → Lake Superior, Two Harbors (-100 ft) | 632,448.627 | 2,109.621539 | 135,015.779 | 234,936.160 | 234,936 | 8.5261 |
| NuMI target → MINOS FD (validation of the target position) | 735,344.614 | 2,452.845607 | 156,982.119 | 256,902.501 | 256,903 | 0.0000 |
| NuMI target → decay-pipe end (TDR design axis) | 724.184 | 2.415619 | 154.600 | 100,074.981 | 100,075 | 0.0197 |

# A History and reproduction of the earlier calculations

The 2013 notebook [3] used the MINOS mast antenna (41°50′23″ N, 88°16′13″ W, 228 m) lowered by 300 m and Ash River antenna 2 (348.638 m); the 2014 note [1] specified 105 + 15 m and 12 + 7.8 m vertical offsets instead and left the horizontal antenna-to-detector vectors to be determined. Table 25 reproduces the notebook's printed outputs with the present code, and Table 26 shows the 2014-note ledger. Table 27 traces the ND → FD tick count through the successive geometries: the 2014 antenna-derived endpoints, the Near Detector placed at a quoted 1,015 m from the target with the antenna's transverse position, and the surveyed faces. The surveyed FD face is 58 m horizontally and 8.8 m vertically from the antenna-derived point; the surveyed ND face is 993.35 m from MCZERO, 3.6 m below the axis, not 5.1 m above it.

Table 25: Reproduction of the 2013 notebook [3]; the last row is the 2014-note offset geometry.

| quantity | recomputed | notebook | relative difference |
|---|---|---|---|
| 2013 notebook, spherical chord (m) | 810,429.1726 | 810,429.1726 | $7.2 \times 10^{-16}$ |
| 2013 notebook, WGS84 chord (m) | 809,606.1622 | 809,606.1622 | $5.8 \times 10^{-16}$ |
| 2013 notebook, ToF (µs) | 2,700.555470 | 2,700.555470 | $6.7 \times 10^{-16}$ |
| 2013 notebook, ToF (ticks) | 172,835.5501 | 172,835.5501 | $5.1 \times 10^{-16}$ |
| 2013 notebook, MI shift (ticks) | 99,920.3817 | 99,920.3817 | $1.5 \times 10^{-16}$ |
| 2013 notebook, total (ticks) | 272,755.9318 | 272,755.9318 | $2.1 \times 10^{-16}$ |
| 2014 note offsets, WGS84 chord (m) | 809,616.2634 | – | – |

Table 26: Correction ledger of the superseded 2014-note geometry (antenna fixes with vertical offsets).

| step | applied | $L$ (m) | $\Delta L$ (m) | $\Delta t$ (ns) | $\Delta$ticks |
|---|---|---|---|---|---|
| reference model | spherical Earth, R = 6378137.000 m; heights as given; surveyed points, no offsets | 810,448.173 | 0.000 | 0.00 | 0.000 |
| Earth model | WGS84 ellipsoid; heights as given; surveyed points, no offsets | 809,625.125 | −823.048 | −2,745.39 | −175.705 |
| vertical offset | ND endpoint if it were under the MINOS mast (105 + 15 m down): endpoint -120.000 m along the local geodetic vertical (105 m cavern depth + 15 m mast) | 809,617.527 | −7.598 | −25.35 | −1.622 |
| vertical offset | Ash River Ant-2 fix, front face via -19.8 m (antenna-derived, superseded): endpoint -19.800 m along the local geodetic vertical (12 m antenna → detector top + 7.8 m to mid-height) | 809,616.263 | −1.263 | −4.21 | −0.270 |

Table 27: ND → FD tick count through the successive geometries.

| geometry | ticks | vs surveyed |
|---|---|---|
| 2013 notebook: antenna −300 m, no FD offset | 172,835.550 | 19.515 |
| 2014 note: antenna −120 m, FD antenna −19.8 m | 172,837.706 | 21.671 |
| ND at the quoted 1015 m, antenna transverse | 172,811.637 | −4.398 |
| surveyed faces (this note) | 172,816.035 | 0.000 |

## B Software and reproducibility

All numbers were produced by `gps-tof` [2] (Python, standard library plus PyYAML) from the configuration `config/nova.yaml`, which records every input with its source. The package implements Eqs. (4)–(11), the correction ledger, the sensitivities and the error budget, and is covered by a test suite that pins the reproduction of the 2013 notebook, the EPSG worked example, the Beams-doc-1148 control points, the survey coordinates and the headline results. To regenerate the report and this note's tables:

```
gps-tof -c config/nova.yaml -f all -o reports/nova
python3 scripts/make_note_tables.py
cd docs/note && latexmk -pdf nova-tof-note
```

Local copies of every cited document are kept in `docs/external/` and `docs/reference/` of the repository, with SHA-256 checksums and an index of provenance.

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